NAMRAA PATEL

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Codeforces Expert (1731) | **LeetCode (2207)**

Education

Pandit Deendayal Energy University (PDEU) — Gandhinagar, India
B.Tech – Computer Science & Business Systems (CSBS), CGPA: ***9.61/10*** — *Expected: May 2028*

Government Polytechnic Ahmedabad — Ahmedabad, India
Diploma Engineering – Computer Science, CGPA: ***9.78/10*** — *Aug 2022 – Apr 2025*

Open Source & Systems Work

Open Source Contributor — Apr 2026 – Present
Celery, Microsoft Agent Framework, and Open Food Facts — *Remote*

- Traced a shared-state mutation bug in Celery's `autoretry_for` decorator to a mutable default `retry_kwargs` dict aliased across task retries (celery/celery #10461); wrote a minimal reproduction, fixed it, and flagged an uncovered edge case in the `getattr` fallback branch during review.
- Tracked down a memory leak in Celery's hard-timeout path to an ineffective `traceback_clear(exc)` call that was silently failing because it targeted a frame still on the call stack (celery/celery #10493). Removed the dead call, relied on `exc.__traceback__ = None` to break the exception–traceback–frame reference cycle, and added a regression test plus a hard-timeout smoke test; merged into the **5.7.0** milestone.
- Resolved a re-entrant-lock deadlock in Celery's Redis result backend by guarding a redundant `UNSUBSCRIBE` call that could re-enter redis-py's non-reentrant `PubSub` lock (celery/celery #10497); added regression tests and merged into **5.7.0**.
- Closed a time-of-use gap in Celery's certificate verification where a certificate could expire mid-run yet still pass signature checks (celery/celery #10510); added an explicit expiry check ahead of verification, plus regression tests. Merged into **5.7.0**.
- Patched an input-mutation bug in Microsoft's Agent Framework where `SerializationMixin.from_dict()` could unexpectedly mutate nested dictionaries in the caller-provided input when dictionary-shaped dependencies were merged during deserialization (microsoft/agent-framework #7901); replaced in-place updates with non-mutating merges and added regression tests. **Merged**.

Django Backend Intern — Jun 2024 – Aug 2024
Techaronic IT Solutions Pvt. Ltd. — *Ahmedabad, India*

- Optimized Django REST APIs with database indexing and ORM query tuning, cutting average API response time by **25%** under production load.
- Implemented authentication/authorization workflows with request validation and access control across application modules.
- Deployed and maintained services on AWS EC2, handling environment configuration and production debugging under live traffic.

Projects

Limit Order Book & Matching Engine | Matching-Engine | *C++20, CMake, GoogleTest, Google Benchmark, TSan/ASan*

- Developed a price-time-priority C++20 matching engine (limit/market, partial fills, cancels, multi-instrument); swapped a `map`/`deque` book for sorted-vector + intrusive lists ($O(1)$ cancel): **13.7K → 3.81M ops/s (278x)** on cancel-heavy load.
- Cut hot-path allocation with a chunked freelist object pool (**99.99%** hit rate): **+22–49%** throughput, up to **5x** lower variance; profiling-driven tuning added **+15%**, with regressing experiments reverted.
- Architected a lock-free SPSC ingestion queue (acquire/release ordering, single-writer matching thread); fixed a real memory-ordering race found by ThreadSanitizer, and caught a silent-drop flaw in my own benchmark, re-validating with submitted = processed checks; **490** test runs under ASan/UBSan/TSan.

TeamSense – Async Task Pipeline for HR Analytics | TeamSense | *Django, Celery, Redis, Docker, React*

- Built an async Celery-Redis task pipeline to move compute-heavy feedback processing off the request cycle, cutting synchronous processing load by **60%**.
- Designed a multi-entity PostgreSQL schema (employees, meetings, feedback) and containerized the stack with Docker for reproducible dev/prod environments.

Personal Cloud Assistant | PersonalCloudAssistant | *React, AWS Lambda, API Gateway, DynamoDB, S3, Cognito*

- Engineered a serverless notes-and-files backend on AWS Lambda and API Gateway, backed by DynamoDB and S3 with presigned download URLs; secured with Cognito JWT authentication and least-privilege IAM roles.
- Instrumented CloudWatch custom metrics for operation counts, request duration, and errors behind an admin-only monitoring dashboard, and shipped a React 19 frontend with Amplify CI/CD deployment.

Technical Skills

Core CS: Data Structures & Algorithms, Concurrency & Correctness, OOP, DBMS, Operating Systems, Computer Networks, System Design
Languages: Python, Java, C++, JavaScript, TypeScript, PHP
Backend & Databases: Django, REST APIs, Flask, FastAPI, PostgreSQL, MySQL, MongoDB, Redis
Cloud & Systems: AWS (EC2, Lambda, S3, DynamoDB, IAM), Docker, Kubernetes, Linux, Git
AI & ML: RAG, Transformer Models, TensorFlow, Hugging Face, Model Evaluation

Achievements

- **JPMorgan Chase Code for Good Winner:** won the 2026 hackathon with a production-ready nonprofit solution.
- Round 2 in **Google The Big Code 2026**; Top 2,500 of 30,000+ in **Amazon ML Challenge 2025**.
- **Open Source Hackathon Winner:** delivered production-ready features under competitive time constraints.
- **Hackout 2025 Finalist:** reached finals on technical merit; OCI Generative AI Professional & Data Science Professional.